OLS Estimator via Singular Value Decomposition

This solution originated as a multiple choice problem from Linear Algebra preparatory work for the MS in Data Science & AI program at USF, beginning summer 2026. What began as selecting the correct answer evolved into a more rigorous proof and learning exercise, with careful attention to the underlying linear algebraic properties and common points of confusion that a terse multiple choice format does not surface.

Problem

Given the OLS regression model $Y = X\beta + \epsilon$, where $X \in \mathbb{R}^{n\times p}$ with $n > p$, $\beta \in \mathbb{R}^{p\times 1}$, and ϵ is mean-zero noise, the OLS estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

Express $\hat{\beta}$ in terms of the SVD of X, where $X = UDV^T$.

Background

Key Result (Orthonormal Vectors)**.** *A set of vectors $\{v_1, \ldots, v_k\}$ is* ***orthonormal*** *if*

$$v_i^T v_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

That is, every vector has unit length, and distinct vectors are mutually perpendicular. The inner product $v_i^T v_j$ measures the projection of v_i onto v_j; orthonormality says each vector has no component in any other direction in the set. This is the natural generalization of the standard basis vectors $e_1, \ldots, e_k$.

Key Result (Orthonormal Matrix — Square Case)**.** *A square matrix $M \in \mathbb{R}^{k\times k}$ is* ***orthonormal*** *(orthogonal) if its columns form an orthonormal set. Writing out the column inner products in matrix form, this is equivalent to*

$$M^T M = MM^T = I_k.$$

This has a powerful consequence for inverses: comparing with the definition $MM^{-1} = I_k$, we immediately read off that $M^{-1} = M^T$. In other words, for an orthonormal square matrix, the inverse is free to compute — it is simply the transpose. A further consequence used below is that $(M^T)^{-1} = (M^{-1})^T = (M^T)^T = M$, i.e. the inverse of the transpose is the original matrix.

Key Result (Orthonormal Matrix — Tall Case)**.** *For a tall matrix $M \in \mathbb{R}^{n\times p}$ with $n > p$ and orthonormal columns, the column orthonormality condition gives only the weaker one-sided identity*

$$M^T M = I_p.$$

The product MM^T is $n \times n$ but has rank at most $p < n$, so $MM^T = I_n$ is impossible — I_n has full rank n, while MM^T can never be. Geometrically, MM^T acts as a map from $\mathbb{R}^n$ to $\mathbb{R}^n$, but its image is only the p-dimensional column space of M — a proper subspace of $\mathbb{R}^n$. Any vector orthogonal to that subspace is sent to zero, so MM^T cannot act as the identity on all of $\mathbb{R}^n$. The one-sided identity $M^T M = I_p$ is nonetheless sufficient for our purposes: it allows us to cancel $M^T M$ to I_p whenever this product appears, which is the key cancellation exploited in Steps 1 and 3 below.

Key Result (Diagonal Matrices and Their Inverses)**.** *A diagonal matrix* $D = \mathrm{diag}(\delta_1, \ldots, \delta_p)$ *with all positive diagonal entries is invertible, with*

$$D^{-1} = \mathrm{diag}(\delta_1^{-1}, \ldots, \delta_p^{-1}).$$

This follows directly from the definition: $DD^{-1} = \mathrm{diag}(\delta_1 \cdot \delta_1^{-1}, \ldots) = I_p$. *A useful consequence is that for any integer power* k, $D^k = \mathrm{diag}(\delta_1^k, \ldots, \delta_p^k)$, *so in particular* $(D^2)^{-1} = D^{-2}$ *and* $(D^{-2})D = D^{-1}$, *mirroring the familiar rules of scalar exponents. Note also that diagonal matrices are symmetric (*$D^T = D$*) and commute with one another.*

Key Result (Inverse of a Product)**.** *For square invertible matrices* A, B:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

This reversal of order is necessary: AB *first applies* B *then* A*, so to undo it one must first undo* A *(apply* A^{-1}*) and then undo* B *(apply* B^{-1}*). The rule extends to three factors as* $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$*, which is applied in Step 2 below. Crucially, this rule requires all matrices involved to be* square *and invertible — it cannot be applied to* X *directly since* X *is not square.*

SVD Components

In the thin SVD $X = UDV^T$:

- $U \in \mathbb{R}^{n \times p}$ has orthonormal columns, so $U^T U = I_p$ (tall case above).
- $D \in \mathbb{R}^{p \times p}$ is diagonal with positive singular values $\delta_1, \ldots, \delta_p$, hence invertible and $D^T = D$.
- $V \in \mathbb{R}^{p \times p}$ is square and orthonormal: $V^T V = VV^T = I_p$, so $V^{-1} = V^T$ and $(V^T)^{-1} = V$.

Derivation

Step 1: Compute $X^T X$ using the SVD

Substituting $X = UDV^T$ and using $D^T = D$ (diagonal matrices are symmetric):

$$X^T X = (UDV^T)^T(UDV^T) = VD^T U^T \cdot UDV^T = VD\underbrace{(U^T U)}_{=I_p}DV^T = VD^2V^T.$$

The cancellation $U^T U = I_p$ is the tall-case orthonormality of U: the columns of U are orthonormal, so their pairwise inner products collapse to the identity.

Remark. *A natural first instinct is to attempt to simplify* $(X^T X)^{-1}X^T$ *by writing* $(X^T X)^{-1} = X^{-1}(X^T)^{-1}$*, and then cancelling* $(X^T)^{-1}$ *against* X^T *to obtain* X^{-1}*. However, this reasoning is invalid because* X^{-1} *does not exist for a non-square matrix. The inverse of a matrix* A *is defined as the matrix* B *satisfying* $BA = AB = I$ *with the same identity matrix on both sides. For a tall matrix* $X \in \mathbb{R}^{n \times p}$ *with* $n > p$*, this is impossible:* XB *is* $n \times n$ *and has rank at most* $p < n$*, so* $XB = I_n$ *can never hold for any matrix* B*. The resolution is precisely Step 1 above: rather than inverting* X *directly, we first form* $X^T X$*, which* is square *(*$p \times p$*) and invertible, and only then apply the inverse-of-a-product rule legitimately.*

Step 2: Invert $X^T X$

Since V is square orthonormal and D^2 is diagonal with positive entries, all three factors in VD^2V^T are square and invertible. Applying the inverse-of-a-product rule to three factors and using $V^{-1} = V^T$ and $(V^T)^{-1} = V$:

$$(X^T X)^{-1} = (VD^2V^T)^{-1} = (V^T)^{-1}(D^2)^{-1}V^{-1} = VD^{-2}V^T.$$

Step 3: Compute $(X^T X)^{-1} X^T$

Substituting $X^T = (UDV^T)^T = VDU^T$ and applying the result of Step 2:

$$(X^T X)^{-1} X^T = VD^{-2}V^T \cdot VDU^T = VD^{-2} \underbrace{(V^T V)}_{= I_p} DU^T = VD^{-2}DU^T = VD^{-1}U^T,$$

where $V^T V = I_p$ is square-case orthonormality of V, and $D^{-2}D = D^{-1}$ follows from the diagonal exponent rule.

Step 4: Conclude

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T Y = VD^{-1}U^T Y.}$$